

INDUSTRIAL APPLICATION

Estimating a Rail Passenger Trip Origin-Destination Matrix Using Automatic Data Collection Systems

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Abstract: Automatic data collection (ADC) systems are becoming increasingly common in transit systems throughout the world. Although these ADC systems are often designed to support specific fairly narrow functions, the resulting data can have wide-ranging application, well beyond their design purpose. This article illustrates the potential that ADC systems can provide transit agencies with new rich data sources at low marginal cost, as well as the critical gap between what ADC systems directly offer and what is needed in practice in transit agencies. To close this gap requires data processing and analysis methods with support of technologies such as database management systems (DBMS) and geographic information systems (GIS). This research presents a case study of the automatic fare collection (AFC) system of the Chicago Transit Authority (CTA) rail system and develops a method for inferring rail passenger trip origin-destination (OD) matrices from an origin-only AFC system to replace expensive passenger OD surveys. A software tool is developed to

facilitate the method implementation and the results of the application in CTA are reported.

1 INTRODUCTION

Public transport agencies have traditionally been hampered in planning, managing and evaluating their services by having to rely heavily on costly and unreliable manual data collection systems. For example most agencies have relied on passenger surveys to estimate origin-destination (OD) travel patterns for current customers. The resulting data is subject to bias and error and is expensive and time consuming both to collect and to process (Attanucci and Wilson, 1981). However, over the past several decades, a number of automatic systems have been introduced into the public transport sector. These systems, including automatic vehicle location (AVL), automatic passenger counting (APC), and automatic fare collection (AFC), are changing the amount, type, and quality of data available to planners and managers.

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While these automatic data collection (ADC) systems are often designed to support specific, narrow functions, it is clear that the resulting data can have wide-ranging application within transit agencies. This article illustrates the potential, as well as the difficulties, of utilizing these systems and demonstrates that, at low marginal cost, ADC systems can provide transit agencies with new data sources to support decision making. The gap between what ADC systems directly offer and what is needed in practice by transit agencies requires extensive data analysis with support of technologies such as database management systems (DBMS) and geographic information systems (GIS) before the full value of the data source can be exploited. This research is an initial attempt to demonstrate this potential by presenting a case study of ADC systems at the Chicago Transit Authority (CTA).

Section 2 summarizes the common characteristics of ADC systems and reviews the state of practice in several transit agencies with operational ADC systems. Section 3 introduces several CTA examples to illustrate the obstacles to be overcome and the techniques required to capitalize on the ADC data. Section 4 examines in detail the task of estimating the OD matrix for current CTA rail passengers. The case study proposes a method for inferring the rail OD matrix from an origin-only AFC system. The proposed algorithm takes advantage of the pattern of a person's consecutive transit trip segments. In particular, the study examines the rail-to-bus transfer case, which is ignored in prior studies (Barry et al., 2002; Rahbee and Czerwinski, 2002), by integrating AFC and AVL data using GIS, DBMS, and other computer-based techniques. A software tool is developed to implement the algorithm and the results of the algorithm application to the CTA are reported. Finally, Section 5 presents the conclusions and explores the potential future use of these ADC systems.

2 CHARACTERISTICS OF ADC SYSTEMS

For the purpose of discussing the general types and typical characteristics, ADC systems are classified as AVL and tracking systems, AFC systems, and APC systems (Brown and Casey, 2000; Casey, 1999; Wile, 2003). Automatic vehicle location and tracking systems are systems that record vehicle location information, which can be used to keep the driver informed of schedule deviation, to inform the control center, and to drive automatic stop announcement systems. Automatic fare collection systems include a wide range of technologies which record the location and time of each customer's fare transactions. Automatic passenger counting systems are used to count passengers boarding and alighting by stop.

2.1 General characteristics of ADC systems

Automatic data collection systems share several common characteristics when compared with traditional manual data collection systems:

Spatial and temporal coverage: ADC systems offer extensive spatial coverage of a transport system, in contrast to traditional methods, which usually focus only on key locations. ADC systems also have a full temporal span, providing continuous data, in contrast to traditional methods in which a "typical" day and peak hours are chosen. Thus, periods that are very infrequently observed with traditional manual data collection, such as weekends, special events, and days with unusual weather can be monitored just as effectively as more representative days.

Data processing and accuracy: ADC data are automatically collected and digitally stored. With the support of an effective data processing and analysis framework, ADC data can be processed very fast. The accuracy of ADC data can be determined, avoiding all sorts of hard-to-detect survey errors and bias that can appear in traditional methods.

Cost structure: ADC systems typically involve high initial investment as well as time to get the system installed, debugged and fully operational, compared with conventional manual collection systems, however, once operational they have relatively low incremental cost for additional data collection.

Inherent disadvantages: Many ADC systems were not initially designed for data collection. When they are looked at as data collection devices, not surprisingly, many limitations arise: some lack critical information, some collect data for only a limited purpose, such as revenue accounting, although they could have been simultaneously collecting data for planning and operations analysis. The data collected are often intermittent, fragmented, and not in an easy-to-use format, which can render the raw data meaningless without extensive processing. Different systems supplied by different vendors often have incompatible data structures and the data are often stored and managed in different DBMS. All of this makes integration across systems difficult and has seriously limited the use of ADC data.

Examples of the applications of ADC systems in the public transit agencies include San Francisco Bay Area Rapid Transit (BART) (Buneman, 1984), several agencies in the Netherlands (Muller and Furth, 2001; Furth et al., 2006) and Portland Tri-Met (Furth et al., 2006; Kimpel et al., 2004; Strathman et al., 2002; Strathman et al., 2002).

2.2 The CTA ADC systems

The CTA is the United States second largest transit system, delivering 1.5 million rides on an average weekday

in Chicago and 38 suburbs. There are three ADC systems currently operating in the CTA: an AFC system, an AVL system, and an APC system. This section focuses on the AFC and AVL systems, introducing the aspects that help illustrate the characteristics introduced in Section 2.1 as well as those related to the case study of estimating the rail passenger trip OD matrix.

2.2.1 The AFC system.

Entry only: CTA's AFC system is an entry-only system, meaning passengers only use their farecards when entering a rail station or boarding a bus, and so no information about a trip's destination is directly provided. Because the CTA fare structure is flat, irrespective of the travel distance, travelers are charged only when entering. Because the AFC system is designed only for fare collection and revenue accounting, it is not concerned with exiting from the system, and thus does not record destination information, although this would be very valuable for transit planning and operations analysis.

Record description: A transaction record is generated each time a farecard is read on the AFC equipment and contains the time, location, event, and unique serial number of that farecard. The time is recorded to the second for each use; the event field distinguishes between trip transactions such as entering a station or boarding a bus, and nontrip transactions such as card purchases, or adding value; and the unique serial number, which is assigned to each farecard, allows tracking an individual rider's multiple trips over the lifetime of the farecard.

Location information: The location is recorded differently for rail trips than for bus trips in the AFC system. For rail, the location is recorded as the station entrance code, whereas for bus, the location is coarsely represented by the bus route number, rather than the exact boarding location (or bus stop ID).

There are fare media other than the AFC system, such as cash payment. In 2004, the AFC system accounted for 88% of the total CTA users.

2.2.2 The AVL system. The AVL system in CTA is part of the automatic voice annunciation system (AVAS), which was introduced by CTA to comply with the Americans with Disabilities Act by providing route and stop information to customers. The AVAS system provides bus location along with the bus identification and status information. The system records a number of events related to vehicle progress along a route such as "stop arrival" and "stop departure." Each event has a timestamp, also to the second, and the location information derived from the dead-reckoning enhanced GPS, as well as identification information such as bus route number and bus number. The location information is recorded as the route number, route pattern ID and the stop sequence number.

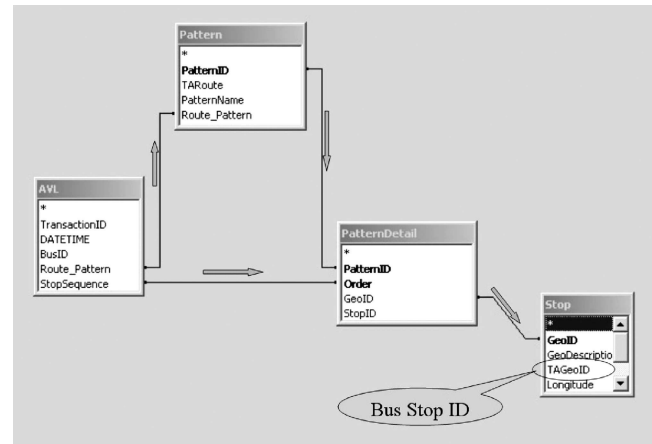


Fig. 1. AVL data processing. Identifying bus stop ID.¹

3 ADC DATA PROCESSING AND INTEGRATION ACROSS SYSTEMS

We will use several practical examples from the CTA to illustrate that although there is great potential for these new ADC data sources, there is a critical gap between what ADC data directly offers and what is desired by transportation planners and analysts, and we show techniques that can be used to bridge this gap. The complexity of the data manipulation in the examples to be presented increases sequentially—from the processing of data from an individual ADC system to integration across multiple ADC systems.

3.1 Processing of data from individual ADC systems

3.1.1 AVL data processing—Identifying bus stop ID. The bus stop ID would be the most useful means of recording the bus location, but the AVL system at CTA does not record it directly in the raw AVL data; instead, it reports only bus route number, route pattern, and stop sequence number. Figure 1 is the entity-relationship diagram that illustrates the AVL data table, the auxiliary tables (pattern, pattern detail, and stop), and their interrelationships. The AVL table is created daily, whereas the auxiliary tables are static. With DBMS and the relationships identified, it is straightforward to translate the route, pattern, and stop sequence into a bus stop ID. In fact, such clear interrelationships as shown in Figure 1 were not documented in the AVL data systems. The difficulty of filtering out the relevant relationships from the dozens of tables and their hundreds of possible inter-relationships without a well-documented AVL data system structure delayed its use considerably.

¹This, and the following, entity-relation diagrams are screen captures from MS Access, however, they are illustrative only. The actual implementation is programmed in C/C++, as discussed in Section 4.2.

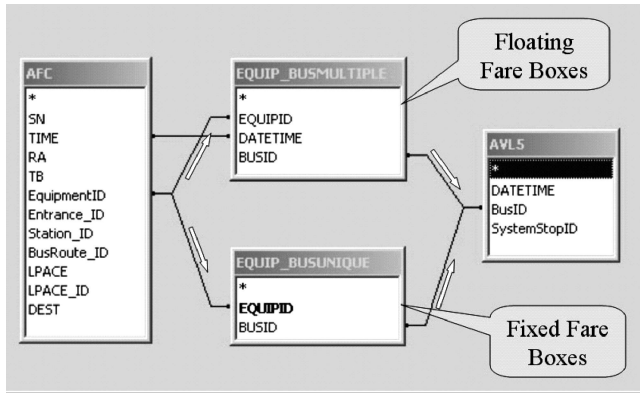


Fig. 2. AFC data processing. Identifying the bus ID.

3.1.2 AFC data processing—Identifying the bus ID. On CTA buses the AFC system records the bus route number but not the bus ID, which would be of greater value in fleet management. Figure 2 shows the technique that resolves this problem by taking advantage of the relationship between the AFC fare box equipment ID and the bus ID. Most fareboxes are installed on specific buses for long periods, but some “float” among buses over time, for example, because of periodic maintenance needs for both buses and fareboxes. In the DBMS, two-table joining operations are performed: one for the fixed fare boxes, for which a simple one-to-one relationship is sufficient; the other for the floating fare boxes, for which the time stamp is needed in addition to the equipment ID to complete the join.

Both these simple examples show that detailed knowledge of the ADC system data structure and data processing in DBMS are required to make any real use of ADC data.

3.2 Integration between the AFC and AVL systems

The CTA AFC system records the bus trip boarding location only coarsely: at the bus route level. More specific boarding location, that is, the bus stop, is necessary to carry out many analyses on passenger bus trips such as bus passenger trip OD analysis. The integration of the AFC system data, which includes trip transactions, and

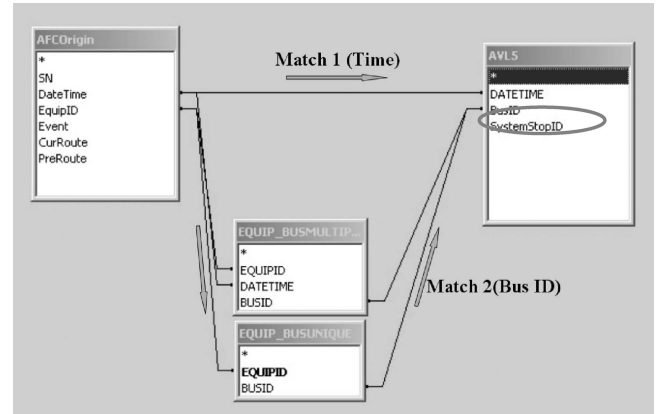


Fig. 3. Combining AFC with AVL to get the boarding bus stop ID.

the AVL system data, which includes vehicle locations, offers a solution: the vehicle location information can be matched against the passenger trip information to infer the passenger’s boarding bus stop ID. The integration of AFC and AVL is realized through two relationships. First, the time when a passenger’s farecard is read should be shortly after the corresponding AVL transaction time. Second the bus IDs from the AFC and AVL must match, as shown in Figure 3.

For example, in Figure 4a, the AFC records two consecutive transactions on farecard 1140031645, showing bus 4521 being boarded at 2:20:05 p.m. In Figure 4b, the three records for bus 4521 indicate that this bus passed stop A at 2:08:00 p.m., stop B at 2:20:00 p.m., and stop C at 2:35:00 p.m. After matching the AFC table with the AVL table, it is reasonable to infer that the passenger boarded at stop B at about 2:20 p.m.

Figure 5 illustrates the full data manipulation process transforming the AFC data from the raw format (indicated by the leftmost table) to data that public transit planners can make use of (indicated by the rightmost table), particularly as input to the passenger OD matrix inference, which will be described in Section 4. The process combines the above examples of processing individual AFC data to get the bus ID, processing individual AVL data to get the stop ID, integrating AFC and AVL

(a) AFC Transactions				(b) Consecutive AVL Records		
Card SN	Time	Type	Origin Location	BUSID	TIME	Bus Stop
...	...	...	...	...	...	...
1140031645	1:20:00 PM	Subway	Station ID 1220	4521	2:08:00 PM	A
1140031645	2:20:05 PM	Bus	(Bus ID 4521)	4521	2:20:00 PM	B
...	...	...	...	4521	2:35:00 PM	C
...	...	...	...	...	...	...

Fig. 4. Example of AFC and AVL combination.

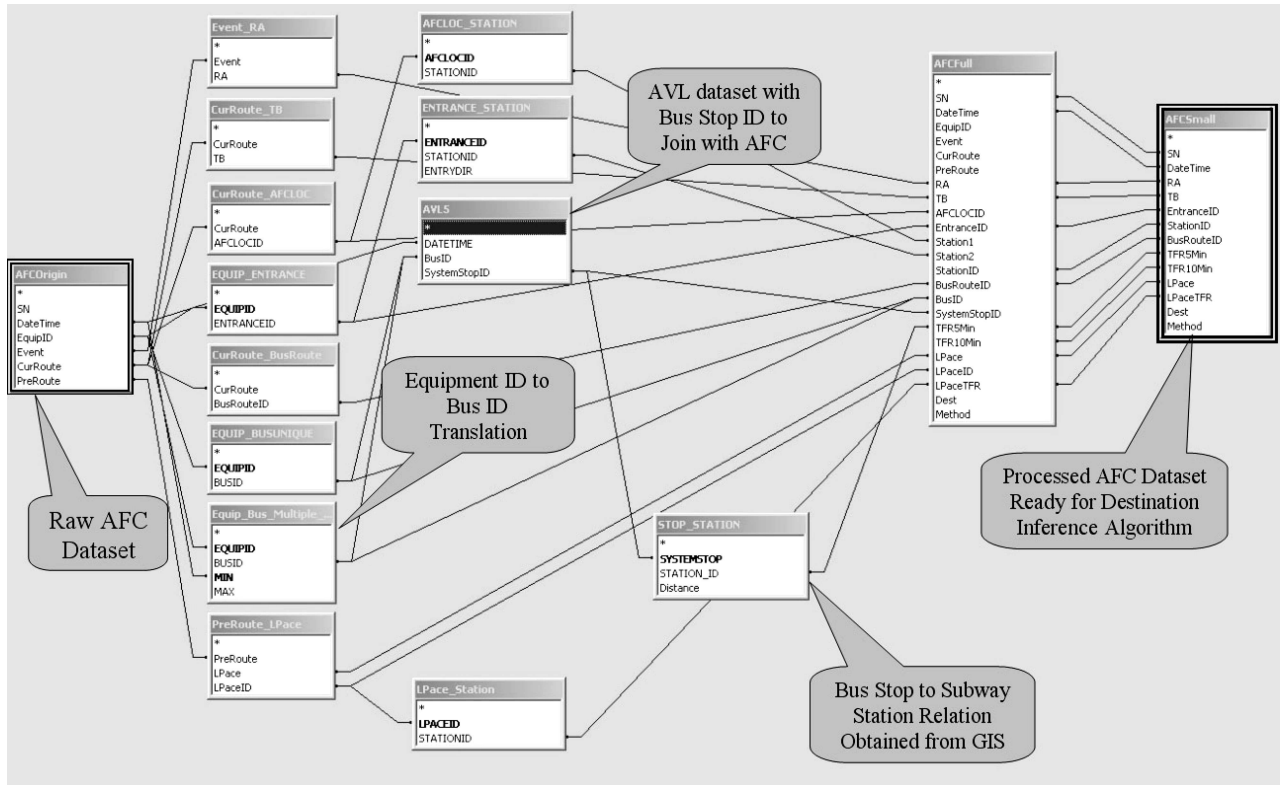


Fig. 5. Full-scale AFC and AVL data processing.

data to get passenger boarding bus stop ID, GIS analysis of the proximity between bus stops and rail stations, and other auxiliary steps.

In the spatial proximity analysis, first both bus stops and the subway stations are geocoded in the GIS system (ESRI ArcGIS v8); second the walking distances between bus stops and rail stations are calculated using the ArcGIS network analyst tool; finally the 1,320 feet (5-minute walk time) is chosen as the threshold to define the reasonable transfer connections between bus stops and rail stations. The spatial relationship between bus stops and rail stations obtained from the GIS analysis is recorded in the “STOP_STATION” table in Figure 5.

This data manipulation process indicates a striking gap between what ADC systems directly offer and what transit agencies need. A strong foundation of data processing and analysis using DBMS and GIS is needed before the full potential of the ADC data can be realized.

4 CTA RAIL OD MATRIX INFERENCE AND ANALYSIS

The rail passenger OD matrix is of fundamental importance both for long-term rail infrastructure investment planning and for daily operations planning. Trip OD

data are traditionally collected through passenger surveys, which are expensive, subject to unknown bias, and updated only infrequently. The implementation of ADC systems provides a chance to infer trip OD information at low marginal cost. The utilization of ADC data to obtain a trip OD entails different complexity for different ADC system configurations. In transit systems which operate with both entry-control and exit-control AFC systems (such as Bay Area Rapid Transit, Washington Metropolitan Area Transit Authority, and the London Underground), the OD matrix can be directly observed; however, in systems such as CTA and New York City Transit Authority, as well as virtually all bus systems, which have entry-only AFC systems, it is not so easy.

There is a wealth of literature on OD matrix estimation both for highway systems (Cheung et al., 2006; Dixon and Rilett, 2005; Wong et al., 2005; Nie et al., 2005; Chen et al., 2005; Baek et al., 2004; Lo and Chan, 2003; Sherali et al., 2003) and transit systems (Wong and Tong, 1998; Lam et al., 2003; Nguyen et al., 1988; Nuzzolo and Crisalli, 2001). Most transit OD matrix estimation methods are based on aggregate passenger counts. In contrast, our method is based on individual passenger travel information obtained from the AFC and AVL systems. This case study examines the problem of how to infer the destination for a given origin for a passenger traveling in

the CTA rail network, which uses only entry fare control (Zhao, 2004). This is the fundamental element in estimating a complete rail OD matrix. The input to the inference process is the result of the AFC and AVL data processing introduced in Section 3. In this section the destination inference algorithm is proposed, a software tool to automate it described, and the result of its application to the CTA rail system to infer the rail trip OD matrix for a 1-week period presented.

4.1 Destination inference: Two consecutive trip segments

The destination inference method takes advantage of the pattern of a person's consecutive transit trip segments, where a trip segment starts with an AFC transaction and ends either with another AFC transaction or when the passenger leaves the transit system. The basic idea is that a high percentage of riders stay at, or return to, the destination station of their previous trip segment to begin their next trip segment (Barry et al., 2002). This premise is valid as long as the following assumptions hold:

1. There is no private transportation mode trip segment (car, motorcycle, bicycle, etc.) between consecutive transit trip segments in a daily sequence;
2. Passengers will not walk a long distance to board at a different rail station from the one where they previously alighted. This condition is equivalent to the distances between rail stations being longer than typically acceptable walking distance, which in this research is assumed to be 1,320 feet or 5 minutes' walk distance at a speed of three miles per hour;
3. Passengers end their last trip of the day at the station where they began their first trip of the day.

Although not all of these assumptions will be true for every passenger, as long as they are true in the great majority of cases, and where they are not true there is no bias, it is reasonable to expect that the inferred aggregate OD matrix should closely approximate actual travel patterns. The partial validation at the aggregate level will be addressed in Section 4.3.

Because the algorithm uses the following trip segment's origin to infer the prior trip segment's destination, it is necessary to examine consecutive trip segments. Because this case study focuses on rail trips, only rail-to-rail (RR), and rail-to-bus (RB) sequences need be examined. Although the RR sequence has been addressed previously (Barry et al., 2002; Rahbee and Czerwinski, 2002), this study also includes the rail-to-bus sequence by integrating the AFC data with the AVL data in the inference process and utilizing DBMS and GIS technology.

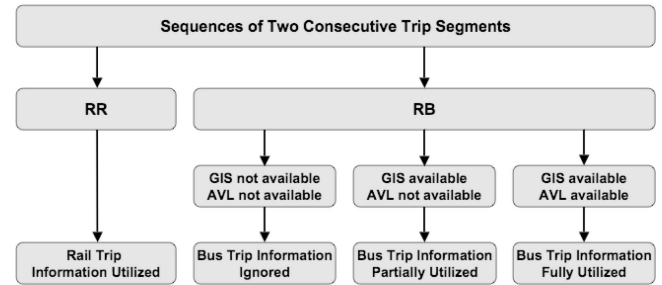


Fig. 6. Cases of two consecutive trip segments.

For rail-to-bus sequences, three contexts are defined in Figure 6 depending on whether AVL and GIS technologies are employed. When neither AVL data nor GIS technology are available, the bus trip segment information cannot be utilized and these trip segment sequences are simply ignored (Barry et al., 2002; Rahbee and Czerwinski, 2002). When GIS technology is employed but no AVL data is available, the bus trip segment information can be utilized in some cases. When both GIS technology and AVL data are available, the bus trip segment information can be utilized to enhance the destination inference algorithm in more cases.

4.1.1 GIS technology without AVL. When GIS technology is employed, the spatial relationship between the rail and bus networks can be analyzed to see if the transfer location can be uniquely identified. Buffers are created as circles around rail stations with 1,320 feet radius. Intersecting these buffers with each bus route determines how many rail stations are served by each bus route. There are three possibilities for connecting between bus routes and the rail network, as shown in Figure 7. In the first case, the bus route does not connect with the rail network, and so the destination of the rail trip cannot be inferred. In the second case when the bus route has only one connection with the rail network, the connection station is identified as the destination of the rail trip segment. In the third case when the bus route intersects the rail network at more than one location the destination cannot be uniquely identified.

4.1.2 Both AVL data and GIS technology available. When both GIS technology and AVL data are available, the boarding bus stop of the bus trip segment can be identified through the integration of AFC and AVL data; and the inferred alighting station of the rail trip segment can be found by using GIS to determine the station closest to a given bus stop. For bus stops with a rail station within walking distance, this station is assigned as the rail segment's destination. For bus stops with no close rail station, the destination of the rail segment cannot be inferred.

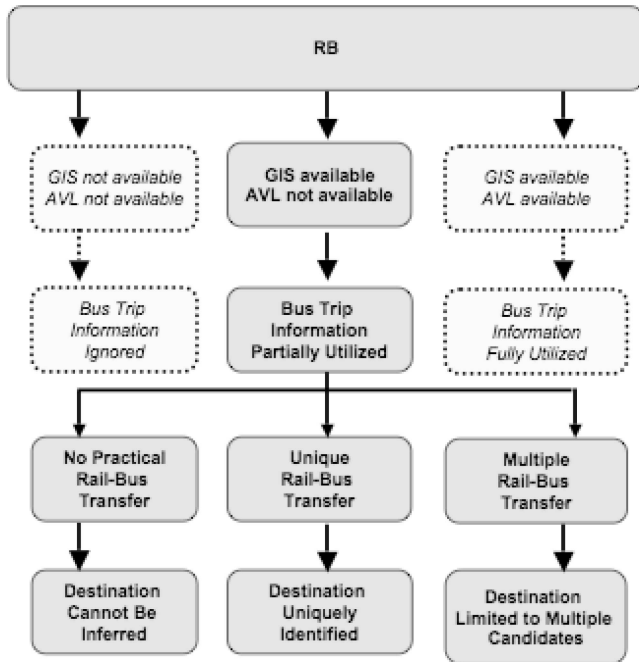


Fig. 7. Scenarios in the rail-to-bus sequence without AVL data.

For example, a person with farecard 1028969715 boarded at Harlem/Lake Station (ID = 20) at 4:54 p.m. and then boarded bus 6205 on route 55 at 5:54 p.m. as shown in Figure 8a. Through the GIS analysis of the spatial relationships between the bus route and the rail network, we know that this person could transfer from the rail network to bus route 55 at any of three connection points—Garfield Station on the Green Line, Garfield Station on the Red Line, or Midway Station on the Orange Line (Figure 9). Without AVL data, it cannot be determined at which station the person actually transferred. However, by combining the AFC and AVL data (Figure 8b), the person can be inferred to have boarded at bus stop 10543. Through the proximity analysis between bus stops and rail stations, we know that Garfield Station on the Red Line (ID = 1,170) is the only rail station close to this bus stop. Therefore, Garfield Station on the Red line is inferred to be the destination of the rail trip segment.

4.1.3 Symmetrical trip chain pattern utilization. In this study, the daily trip segment chain patterns were examined to improve the destination inference algorithm. Certain trip segment chains exhibit symmetry in the full chain, or in part of the chain. For example, in the chain “RBBR,” or any chains that include “RBBR,” the algorithm can check whether the bus route numbers of both bus trip segments are the same to verify symmetry.

If symmetry is confirmed, the later R segment can be used to infer the destination of the first R segment so that the inference is expanded from using only consecutive trip segments to non-consecutive trip segments with symmetrical patterns.

4.2 Development of partial² rail OD inference software

To facilitate this inference process, software was developed which allows someone without knowledge of the details of the ADC systems or in-depth understanding of database and GIS to complete the destination inference process rapidly. Users only need specify the input files and several parameters. This software tool takes the raw AFC and AVL data and several accessory tables as input, and estimates the rail partial OD matrix automatically.

With the AFC data and the AVL data both on the order of millions of records,³ computational efficiency becomes a concern during the implementation, especially when integrating the AVL data with the AFC data. We chose to use C/C++ as the programming language although alternatives such as PL/SQL in Oracle and Transact-SQL in MS SQL Server were also possible. C/C++ does not have the built-in database operating functions of PL/SQL and Transact-SQL, therefore the data indexing, sorting, searching, and table joining had to be written at the primitive levels. A binary search routine is used to match the bus AVL records with the AFC records. But as a result of the low-level access to computer memory via pointers and data structures customized for this particular application, C/C++ offers superb computational performance. The software is compiled within the GNU C/C++ Compiler (GCC) on the MIT Athena Unix system.⁴ The software completed the inference of a six-day trip OD matrix for the CTA rail network in about one hour on a Sun Blade 150 workstation.

4.2.1 Nested loop structure. The algorithm starts by sorting the entire AFC data table using the serial number (SN) of the farecard as the primary key and the transaction date and time as the secondary key so that the trip segments of the same farecard are grouped and sequenced chronologically. As indicated in Figure 10, there are three nested loops in the algorithm. The outer loop

²The term partial OD matrix is used here to refer to the destination inference applied to all AFC rail transactions. The process of expanding this partial matrix to represent all CTA rail customers, including non-AFC customers, is a second stage of the process which will be discussed in Section 4.3.

³The raw AFC table has 5.95 million records and the raw AVL table has 1.62 million records for the six-day period analyzed in January 2004.

⁴The complete C/C++ source code is available at <http://mit.edu/jinhua/www/thesis2004/sourcecode/>

(a) Consecutive AFC Transactions for Farecard 1028969715

Time	Type	Origin Location
...	...	...
4:54:41 PM	Subway	Station ID 20
5:54:27 PM	Bus	Route 55, Bus 6205
...	...	...

(b) AVL Transactions for Bus 6205

BUSID	TIME	Bus Stop ID
...	...	...
6205	5:49:10 PM	14118
6205	5:54:27 PM	10358
6205	5:57:45 PM	3385
...	...	...

Fig. 8. Combining AFC and AVL data.

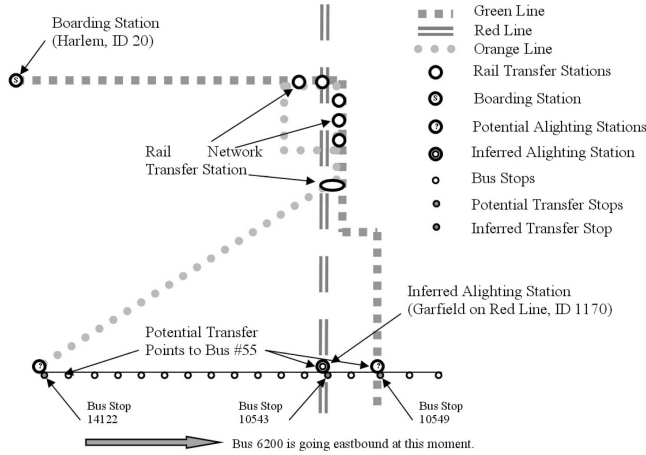


Fig. 9. Bus-rail connections.

cycles through each farecard, the middle loop through each day, and the inner loop through each trip segment of the farecard and the day identified. For each rail trip segment, a procedure is carried out to infer the destination of the trip segment.

4.2.2 Destination inference procedure. Figure 11 shows the overall procedure structure. Suppose the rail trip segment currently under examination is trip segment j by person (farecard) p on day d , and its immediate following trip segment is k . The algorithm determines whether trip segment k is on rail or on bus and moves onto the “RR transfer sub-procedure” or “RB transfer sub-procedure” accordingly except for two special situations: (1) when trip segment j is the last trip segment of the day, a special sub-procedure is called; (2) when a Pace⁵ bus trip segment intervenes before trip segment k , another special sub-procedure is invoked. The RB sub-procedure is described below. The full algorithm is described in Zhao (2004).

⁵Pace is the suburban bus operator in Chicago, which also has an AFC system, but it is not fully integrated into the CTA data system. The “previous Pace sub-procedure” integrates the Pace AFC data with the CTA AFC data.

4.2.3 RB transfer procedure. In the RB procedure, the algorithm first checks whether the bus trip segment is made on an AVL-equipped bus. For AVL-equipped buses, the boarding bus stop is obtained and the closest rail station to the bus stop identified. If the bus stop and rail station are within walking distance, this station is assigned as the destination of the rail trip segment. If the bus stop is beyond a reasonable walking distance from the rail station, then the destination of the rail trip segment cannot be inferred. If there is no valid AVL data for the bus, the algorithm turns to the secondary method for non-AVL-equipped buses. The rail to bus route transfer possibility is examined. If there is only one transfer station from the rail network to the bus route, this station is

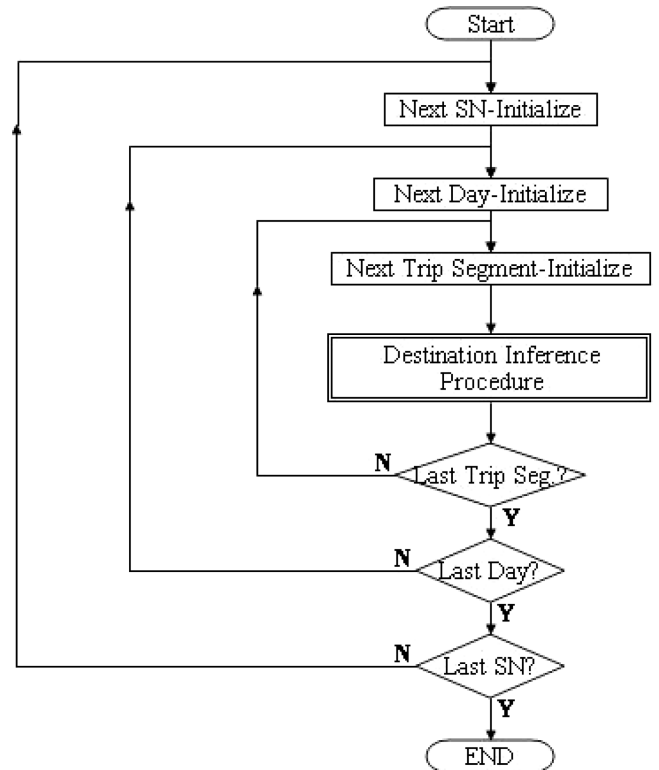


Fig. 10. Nested loop structure of the algorithm.

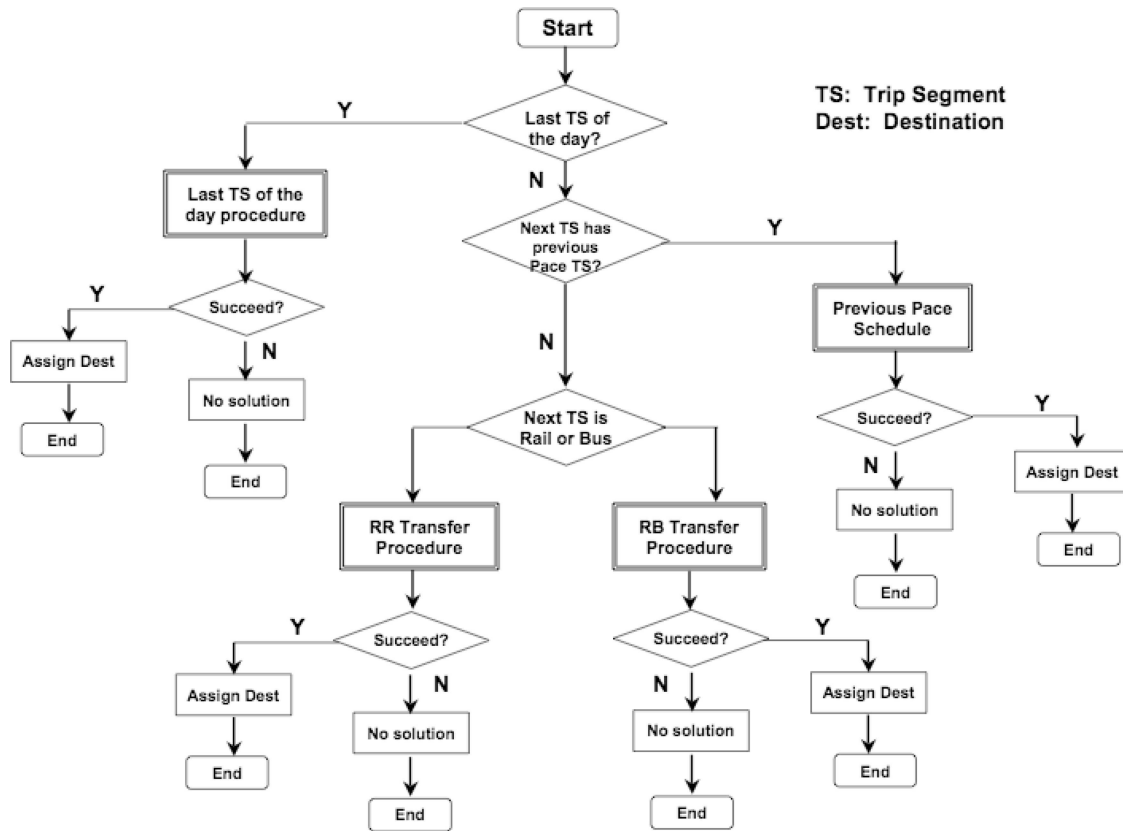


Fig. 11. Procedure for one trip segment's destination inference.

then assigned as the destination of the rail trip segment; otherwise, no destination can be inferred for the rail trip segment (Figure 12).

4.3 Application results and partial verification at the aggregate level

The algorithm was applied initially to the CTA rail system for a 6-day period in January 2004. The CTA AFC and the AVL data were input to the Rail OD Inference Tool and the corresponding rail partial OD matrix produced. There were over 2.5 million rail trip transactions recorded by the AFC system in this 6-day period, of which the algorithm “successfully” inferred the destinations of over 1.8 million, or 71.2%. Table 1 summarizes the contribution of each inference method. The inclusion of the rail-to-bus cases accounts for 14.4% of all destination inferences, which is the value added by the enhancements in this inference method over prior approaches. Of the 28.8% of rail AFC transactions for which a destination was not inferred, a majority (18.6%) are due to there being only a single rail transaction on the AFC card during the day, with most of the remainder being due to no match with either the AVL or GIS methods.

Because the model results in a subset of OD pairs rather than a complete set of all trips, an expansion procedure is needed to estimate a complete OD matrix for all CTA rail trips. This model application results in a 71.2% sample of farecard records with inferred destinations, together with an 88% farecard usage rate, resulting in approximately 64% of all CTA rail trips being captured, reflected in the partial OD matrix which is inferred. The standard iterative proportional fitting method is used to expand the partial OD matrix to the full OD matrix. Two specific implementations are (1) singly-constrained scaling to the known half-hour entries by station, and (2) doubly-constrained scaling to both the known half-hour entries by station and 24-hour exit totals at each station. In the latter case the exit totals are assumed to equal the 24-hour entry totals at the same station.

Potential errors in the estimation and expansion process include:

1. Possible differences in travel behavior between the farecard users and nonfarecard users. Although individually this difference may be significant, we expect that the difference will be smaller at the aggregate level. With the farecard market share

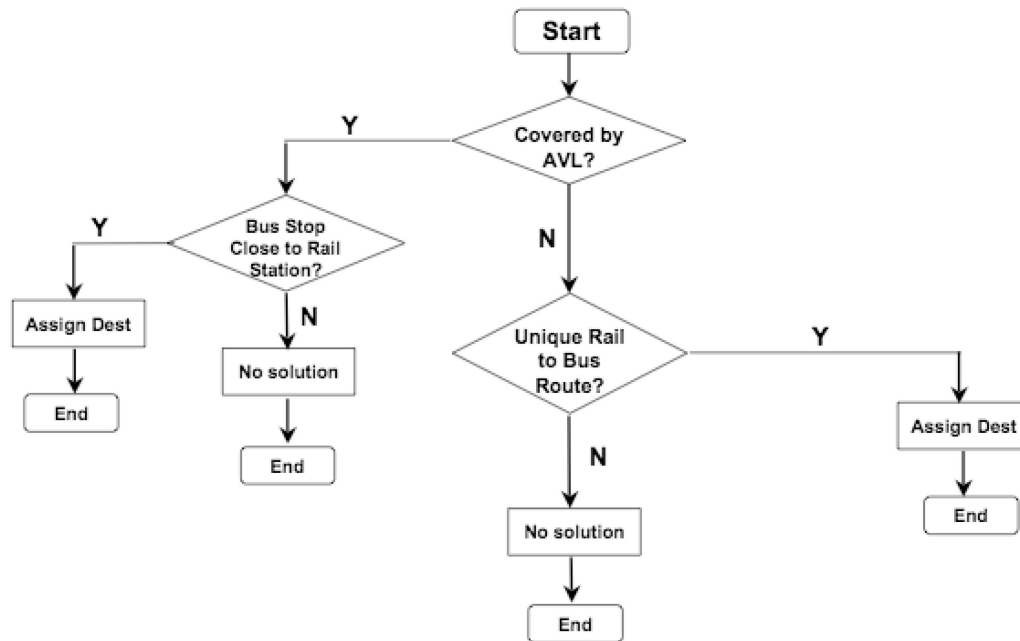


Fig. 12. RB transfer procedure.

growing continuously, it is becoming an even less significant issue. That being said, further research is needed to examine the demographics and geographic distributions of the nonfarecard users to see whether they live in specific areas of the city or are from specific socioeconomic segments, which may introduce bias that should be corrected for in the full OD matrix estimation.

- For the rail AFC transactions for which a destination was not inferred by the algorithm, whether their trip OD pattern differs significantly from the successfully inferred ones, or collectively the travel patterns are similar for the two groups.

Table 1
CTA rail OD inference results

AFC sequence	Method	Total (%)
<i>Contributions of each inference method:</i>		
RR	Next rail trip	33.6
RR	Last rail trip of day	21.1
RB	AVL determination of transfer station	13.0
RB	Unique bus route-rail transfer station	1.4
	Other methods	2.1
INFERRED TOTAL		71.2
<i>Unable to infer because:</i>		
R	Single rail trip	18.6
RB	AVL and GIS failure	5.4
	Other reasons	4.8

- In certain stations, the assumption of the total daily entries equaling the total exits might not hold, for example, at the O'Hare airport station. This will influence the doubly-constrained scaling method in the expansion stage.

Although further work is needed to fully verify the accuracy of the proposed method, a partial verification of the method has been made at an aggregate level. TransInfo was employed by the CTA in 2004 to conduct a statistical test of the aggregate results based on this OD matrix inference method. The resulting report compares the passenger mile traveled (PMT) estimated by this inference method with those obtained from a large CTA customer OD survey and concludes that PMT estimates obtained using the new method meets the Federal Transit Administration (FTA) precision requirements, that is, the (one-sided) width of the 95% confidence interval of the PMT is no greater than 10% of the PMT. FTA has accepted this method as a basis for reporting PMT for the National Transit Database (TransInfo, 2004).

5 CONCLUSIONS

Many ADC systems in public transit agencies have been designed and implemented with single objectives: to announce bus stops, to collect fares, to report emergencies, etc. This article takes another perspective—if each of these systems is looked at as an individual data collection

technology, a new method of data collection exists. Once the ADC systems are installed, they continuously generate huge volumes of data. This article examines the benefits, as well as the difficulties, of utilizing this data, and demonstrates that at very low marginal cost, the increasing use of ADC systems in transit provides the opportunity to obtain a much clearer picture of ridership patterns.

Customer OD matrices are fundamental information required in public transit planning and operations analysis. This research proposed an enhanced method of inferring the rail trip OD matrix from an origin-only AFC system as an alternative to traditional passenger surveys.

The proposed algorithm takes advantage of the pattern of a person's consecutive transit trip segments and uses the next trip segment boarding location information to infer the destination of the prior trip segment. Prior studies examined only the rail-rail trip sequence case and ignored the rail-bus trip sequence case. This article examines the rail-to-bus case by integrating the AFC and AVL system data and examining the spatial connectivity between the bus and rail networks taking advantage of GIS technology. In the CTA rail system application, the inclusion of the rail-to-bus cases account for 14.4% of the destinations inferred, which is the value added by this enhanced inference method.

Other example applications of the ADC systems at CTA include: the analysis of the diverse transit trip segment chain patterns at both the daily and the weekly period; rail path choice behavior modeling using discrete choice method and data from the ADC systems (Zhao, 2004); the analysis of transit access and usage patterns of smart card holders using smart card registration and AFC transaction data (Utsunomiya et al., 2006; Gupta, 2006); and bus passenger OD matrix estimation using ADC systems (Cui, 2006).

To suggest what a future system might be able to contribute, making use of ADC systems which are available today, consider a multimodal public transport system with an AFC system including smart cards with only entry controls on both bus and rail networks, an AVL/APC system for the bus system and an effective train tracking system for the rail network. In addition to the operational performance data which will facilitate more reliable and robust schedules for both bus and rail, with relatively straightforward applications similar to those described above, it should eventually be possible to understand the behavior of public transport customers much better. For example, using the types of inference algorithms discussed in this article, but extended to the full multimodal network, it should be possible to estimate customer OD movements and from there go on to estimation of modal preferences and path choice parameters given the actual and historical quality of service on each link and at each interchange point in the

network. Given AVL data, it should also be possible to characterize the actual service quality on each path for any OD pair at any point in time, including service reliability measures which have previously been infeasible because of the inadequacy and expense of manual data collection. By analyzing the AFC data we can observe a particular individual's travel behavior over time and relate that behavior to the service quality being provided over time on specific paths in the network. By linking system usage to home addresses we can also hope to understand access behavior much better, for instance, we can see how individuals change their behavior with weather or with the impact of improved customer information systems.

These are interesting next steps on the path to taking full advantage of the public transport sector's transition from data-poor to data-rich. This should eventually allow us to understand better both the multidimensional performance of public transport services and its impact on travel behavior. This represents a real sea change in the way that public transport managers and planners can understand their systems and their customers, and we can expect significant changes in the effectiveness and efficiency of public transport service to result in the long run.

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